Inventory Manager — Presentation Outline

> ***Prepared for:*** *PPT / Slide Deck*
> ***Project:*** *Inventory Manager (Full-Stack)*
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Slide 1 — Title Slide

- **Title:** Inventory Manager
- **Subtitle:** A Full-Stack, Multi-Platform Inventory & Retail Management System
- **Tagline:** "End-to-end inventory control — from warehouse to customer, desktop to mobile"
- **Components:** Backend APIs · Desktop App · Mobile PWA

Slide 2 — Problem Statement

- Manual inventory tracking leads to errors, stock-outs, and lost revenue
- Lack of real-time visibility across multiple storage locations
- No integration between purchasing, sales, and inventory operations
- Difficulty tracking profitability at the product level
- No automated alerts for low stock, expiring items, or shelf restocking

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Slide 3 — Solution Overview

- **Unified platform** covering the entire inventory lifecycle
- **Multi-location tracking** — Warehouse, Store, External (FBA)
- **Batch-level control** with expiry dates and FIFO logic
- **Role-based access** for Owner, Manager, Employee, IT Admin, and Customer
- **Multi-channel sales** — In-store POS + Amazon/Flipkart webhooks
- **Cross-platform** — Desktop web app + Mobile PWA with barcode scanning

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Slide 4 — System Architecture

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┌──────────────┐   ┌──────────────┐   ┌──────────────┐
│  Desktop App │   │  Mobile PWA  │   │  Marketplace │
│  (React/MUI) │   │ (Capacitor)  │   │  Webhooks    │
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                  │
                  │ REST API + WebSocket
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        ┌─────────────────┐
        │  FastAPI Backend │
        │  (~100+ endpoints)│
        │  19 Router Modules│
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        │   PostgreSQL    │
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        │   (Docker)      │
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- **Backend:** Python + FastAPI (19 routers, ~100+ REST endpoints)
- **Desktop:** React 19 + TypeScript + Material-UI (MUI)
- **Mobile/PWA:** React + TypeScript + Capacitor (Android APK)
- **Database:** PostgreSQL 15 (Docker)
- **Real-time:** WebSocket for barcode scanner communication

Slide 5 — Technology Stack

Layer	Technology
Backend Framework	FastAPI (Python)
Database	PostgreSQL 15 (Docker Compose)
ORM / DB Access	psycopg2 + SQLAlchemy
Authentication	JWT (OAuth2 password flow) + bcrypt
Scheduler	APScheduler (auto-backup, stock checks)
PDF Generation	ReportLab (A4 GST invoices, reports)
Data Science	pandas, scikit-learn, mlxtend
Frontend Framework	React 19 + TypeScript
UI Library	Material-UI (MUI) v7 + MUI X Data Grid
Charts	Recharts
State Management	Zustand
Build Tool	Vite
Mobile	Capacitor (PWA → Android APK)

Slide 6 — User Roles & Access Control

Feature Area	Employee	Manager	IT Admin	Owner
View Products & Stock	✅	✅	✅	✅
Transfer Stock	✅	✅	✅	✅
Create Sales / POS	✅	✅	✅	✅
Create Products	❌	✅	❌	✅
Receive Stock / Write-Off	❌	✅	❌	✅
Analytics & Profitability	❌	✅	❌	✅

User Management	❌	❌	✅	✅
Backup & Restore	❌	❌	✅	✅
Block Managers	❌	❌	❌	✅

- **5-tier hierarchy:** Owner → IT Admin → Manager → Employee → Customer
- **JWT tokens** with role claims (60-min expiry)
- **Owner protection:** Only owners can manage managers

Slide 7 — Core Feature: Product & Supplier Management

- **Product CRUD** — Create, update, delete products with SKU uniqueness validation
- **Supplier CRUD** — Manage supplier contacts, emails, locations
- **Product–Supplier Linking** — Many-to-many relationships with supply price per link
- **Preferred Supplier** — Mark preferred supplier per product for quick reordering
- **Barcode Support** — Products linked to physical barcodes for scanning
- **Catalog View** — Browse products with stock levels, pricing, and "Add to PO" action

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Slide 8 — Core Feature: Multi-Location Inventory

- **Location Types:** Warehouse, Store, External (FBA)
- **Batch Tracking** — Each stock receipt creates a batch with:
  - Unique batch code (auto-generated)
  - Unit cost, expiry date, received timestamp
- **Stock Operations:**
  - 📥 **Receive** — Single or bulk receive into warehouse
  - 🔄 **Transfer** — Move stock between locations (FIFO-based)
  - 🗑️ **Write-Off** — Damaged, expired, lost stock with reason tracking
- **Weighted Average Cost** — Automatically recalculated on each receive
- **FIFO Logic** — Oldest batches deducted first (by `received_at`)

Slide 9 — Core Feature: Point of Sale (POS) & Billing

- **In-Store Sales** — Create orders with auto FIFO stock deduction
- **Payment Methods:** Cash, Card, UPI
- **Customer Linking** — Associate sales with loyalty customers by phone lookup
- **Profit Tracking** — Cost snapshot at time of sale (`unit_cost` recorded per line item)
- **Receipt PDF** — Generate downloadable receipt for each order
- **Sales History** — Paginated, searchable, filterable order history
- **Export** — Sales list PDF export with filters
- **Barcode Scanning at Checkout** — Scan items directly into the billing page via WebSocket

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Slide 10 — Core Feature: Purchase Order Management

- **Full Lifecycle:** Draft → Placed → Received (or Cancelled)
- **Smart Merge:** Adding items for same supplier auto-merges into existing draft PO
- **Item Management:** Add/remove items, quantity updates, auto-recalculate totals
- **Receive PO:** Converts PO items into inventory batches automatically

- **Draft Check:** Quick lookup of active drafts per supplier
- **"Add to PO" from Catalog:** One-click add from product catalog to draft PO

Slide 11 — Feature: B2B Wholesale Management

- **B2B Client Management** — Full CRUD with credit limits, GSTIN, price tiers
- **B2B Orders** — Wholesale orders with per-client pricing and margin tracking
- **Credit System:**
  - Per-client credit limits
  - Balance tracking (amount owed vs. paid)
  - Auto-block when exceeding credit limit
- **B2B Dashboard:**
  - Total amount to collect
  - Clients over credit limit
  - Top debtors list
- **Frequent Items** — Show client's frequently purchased items with last-sold price
- **B2B Invoice PDF** — A4 GST-compliant invoice generation

Slide 12 — Feature: Khata (B2C Credit Ledger)

- **Customer Credit Accounts** — Name, phone, credit limit, running balance
- **Transaction Ledger** — Complete history of credits and payments per customer
- **Payment Recording** — Cash, UPI, or card payments with reference tracking
- **Auto-Blocking** — Customers exceeding credit limit are automatically blocked
- **Khata Dashboard:**
  - Total credit outstanding
  - Customers with balance, over-limit, near-limit, blocked
- **POS Integration** — Phone lookup at checkout to check credit eligibility
- **WhatsApp Reminders** — Send payment reminders to customers

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Slide 13 — Feature: GST Invoice System

- **A4 Professional Invoice PDF** — Indian GST-compliant format
- **Invoice Types:**
  - Sales Order Invoice (B2C)
  - B2B Invoice (with GSTIN)
  - Purchase Order Invoice (Self-Billed)
- **Customizable Settings:**
  - Business name, address, GSTIN, PAN
  - Custom logo, colors, and footer text
  - Tax rate configuration
- **Amount in Words** — Auto-converts totals to Indian currency words
- **Live Preview** — Preview invoice with settings before saving

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Slide 14 — Feature: Loyalty Points Program

- **Points Earning** — Configurable earn rate (e.g., 1 point per ₹100 spent)
- **Points Redemption** — Redeem points for discount at checkout

- **Customer Lookup** — Phone-based lookup showing points balance
- **Checkout Integration** — Show points preview during billing
- **Settings Management** — Manager can adjust earn rate and redemption value
- **Automatic Points Addition** — Points added after sale completion

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Slide 15 — Feature: Barcode Scanner (WebSocket)

- **Wireless Scanning** — Phone scans barcodes, Desktop receives results in real-time
- **WebSocket Architecture:**
  - Phone connects as `scanner` role
  - Desktop connects as `billing` role
  - Scans broadcast from phone → all connected desktops
- **Two Modes:**
  - 🛒 **Billing Mode** — Scan adds product to checkout
  - 📥 **Receiving Mode** — Scan receives +1 unit into inventory
- **Product Lookup** — Barcode → product details (name, price, stock)
- **Mobile PWA Scanner Page** — Dedicated barcode scanning page in PWA

Slide 16 — Feature: Reports & Exports

Report	Description	Format
Stock Summary	Aggregate stock by product with filters	JSON / CSV / PDF
Location Summary	Stock grouped by location with values	JSON / CSV / PDF
Batch-Wise Stock	Detailed batch-level view with expiry	JSON / CSV / PDF
Physical Register	Blind mode for unbiased stock counting	JSON / CSV / PDF
Low Stock / Reorder	Products below threshold with suggested qty	JSON / CSV / PDF
Stock Ageing	Days since received for each batch	JSON / CSV / PDF
Near Expiry	Products expiring within X days	JSON / CSV / PDF
Overstock / Dormant	No sales for X days (default: 90)	JSON / CSV / PDF
Item Profitability	Revenue, cost, profit per product	JSON / CSV / PDF
Stock Movement	Inbound/outbound/transfer history	JSON / CSV / PDF
Daily Transactions	All operations grouped by day	JSON / CSV / PDF
Supplier Performance	Orders, value, delivery rate per supplier	JSON / CSV / PDF

Slide 17 — Feature: Analytics Dashboard

- **Inventory Valuation** — Total stock value (qty × avg cost)
- **Sales Summary** — Total sales and order count for date range
- **Sales Trends** — Daily sales chart (configurable date range)
- **Top Selling Products** — Ranked by revenue with units sold

- **Write-Off Summary** — By reason (expired, damaged, lost, other) with value
- **Role-Based Dashboards:**
  - 📊 **Manager Dashboard** — KPIs, sales trends, quick actions
  - 👤 **Employee Dashboard** — Shift summary, pending tasks, activity feed
  - 🔧 **Admin Dashboard** — System health, alerts, recent activity

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Slide 18 — Feature: Data Science & AI

- **ABC Classification** (Pareto Analysis)
  - A-items: 80% of revenue
  - B-items: 15% of revenue
  - C-items: 5% of revenue
  - Helps prioritize stock management focus
- **Market Basket Analysis** (Apriori Algorithm)
  - "Frequently bought together" patterns
  - Confidence and lift scores for product associations
  - Uses: mlxtend library
- **Customer Segmentation** (RFM Model)
  - Recency, Frequency, Monetary analysis
  - Segments: VIP, Loyal Active, New Customer, At Risk, Lost, Standard
  - Actionable insights for marketing

Slide 19 — Feature: Alerts & Notifications

- **Automated Stock Alerts** (checked every 10 minutes):
  - 🔔 **Shelf Restock** — Store stock < 5 units
  - ⚠️ **Low Stock** — Total stock < 20 units
  - Auto-resolves when stock replenished
- **Alert Severity Levels:** Info, Medium, Warning, Critical
- **Alert Actions:**
  - Quick shelf transfer from alert
  - Bulk restock (creates purchase order from alert items)
  - Resolve, request closure, confirm fix workflow
- **Issue Reporting** — Users can report bugs/problems visible to IT Admin
- **Notification Badge** — Unresolved alert count on navigation

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Slide 20 — Feature: System Administration

- **Database Backup & Restore:**
  - Manual backup trigger (PostgreSQL dump)
  - Auto-backup daily at 12:00 PM (APScheduler)
  - Auto-delete backups older than 7 days
  - Full restore capability
- **Audit Logs:**
  - Every API action logged with timestamp, user, IP, details
  - Filterable by date range, user, action type, keyword
- **Operations Logs:**

    - Write-offs and backup operations tracked separately
    - Filter by operation type, user, date
- **User Management:**
    - Register, block/unblock users
    - Assign/remove/switch roles
    - Owner protection for manager-level actions

Slide 21 — Feature: External Marketplace Integration

- **Amazon & Flipkart** order ingestion via webhook
- **Endpoint:** `POST /api/v1/integrations/webhook/order`
- **Authentication:** API Key (header-based, not JWT)
- **Fulfillment Methods:**
    - **FBA (Fulfilled by Amazon):** Deducts from External location
    - **FBM (Fulfilled by Merchant):** Deducts from Warehouse → Store
- **Idempotent Processing** — Duplicate `external_order_id` check
- **SKU Resolution** — Maps external SKUs to internal products
- **Profit Tracking** — Records `unit_cost` snapshot for each order

Slide 22 — Multi-Platform Experience

Feature	Desktop App	Mobile PWA
Dashboard	✅ (Manager/Employee/Admin views)	✅
Product Catalog	✅	✅
Inventory Management	✅	✅
POS / Billing	✅	✅
Purchase Orders	✅	✅
B2B Management	✅	✅
Khata (B2C Credit)	✅	✅
Reports	✅	✅
Analytics	✅	✅
Data Science	✅	✅
Barcode Scanner	Receives scans (billing page)	Sends scans (camera)
Invoice Settings	✅ (full config)	✅
Stock Alerts	✅	✅
System Admin	✅	✅
QR Pairing	Generates QR	Scans QR to connect

Slide 23 — Workflow: Stock Receiving

```
1. Manager creates Purchase Order (Draft)
2. Items added to PO (from catalog or manually)
3. PO submitted (Draft → Placed)
4. Goods arrive → PO received
5. System auto-creates inventory batches
6. Weighted average cost updated per product
7. Audit log entry created
```

Alternative: Direct receive via scanner (scan barcode → +1 unit)

Slide 24 — Workflow: Sales / Point of Sale

```
1. Employee opens Billing page
2. Products added via search or barcode scan
3. Optional: Customer phone lookup (loyalty/khata)
4. Payment method selected (Cash / Card / UPI)
5. Order submitted → auto FIFO stock deduction
   - Store batches first (oldest), then Warehouse
6. Cost snapshot saved per item (for profit tracking)
7. Loyalty points earned (if customer linked)
8. Receipt/Invoice PDF available for download
```

Slide 25 — Workflow: B2B Order & Credit

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1. B2B client created with credit limit & GSTIN
2. Wholesale order placed with per-client pricing
3. Order total added to client's outstanding balance
4. If balance exceeds credit limit → client auto-blocked
5. GST invoice generated for the order
6. Client makes payment → recorded in ledger
7. Balance updated, client unblocked if within limit
8. Dashboard shows top debtors and collection analytics
```

Slide 26 — Workflow: Automated Alerts

```
Every 10 minutes (APScheduler):

Check Shelf Stock:
  → Store stock < 5 units? → Create "Shelf Restock" alert
  → Store stock ≥ 5 units? → Auto-resolve existing alert

Check Total Stock:
  → Total stock < 20 units? → Create "Low Stock" alert
  → Total stock ≥ 20 units? → Auto-resolve existing alert
```

```
Employee sees alert → Quick actions:
  → Transfer from warehouse to store shelf
  → Create purchase order for restocking
```

Slide 27 — Key Business Logic Highlights

Logic	Description
FIFO Deduction	Sell oldest batches first (`ORDER BY received_at ASC`)
Weighted Average Cost	`(old_qty × old_avg + new_qty × new_cost) / total_qty`
Profit = Revenue − Cost	Uses actual `unit_cost` snapshot, not average
Smart PO Merge	Auto-merges items into existing draft for same supplier
Alert Auto-Resolution	Stock alerts resolve when thresholds met
Credit Auto-Block	Khata/B2B customers blocked when exceeding limit
Idempotent Webhooks	Duplicate external order IDs rejected

Slide 28 — Security & Compliance

- **Authentication:** OAuth2 + JWT with bcrypt password hashing
- **Role-Based Access Control (RBAC):** 5-tier hierarchy enforced per endpoint
- **Audit Trail:** Every action logged with user, timestamp, IP address
- **Owner Protection:** Critical actions require owner-level privileges
- **GST Compliance:** A4 invoices with GSTIN, tax breakdown, amount in words
- **Database Backup:** Automated daily + manual trigger with 7-day retention
- **API Key Auth:** Separate authentication for external marketplace webhooks

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Slide 29 — Project Stats

Metric	Count
Backend API Endpoints	~100+
Backend Routers	19 modules
Desktop Pages	19
Desktop Components	56+
PWA Pages	20
Database Tables	20+
User Roles	5
Report Types	12
Scheduled Jobs	3

Data Science Models	3 (ABC, RFM, Market Basket)

Slide 30 — Future Scope / Roadmap (Optional)

- ~~Mobile app development~~ (PWA with Capacitor ✅)
- Multi-store chain management
- Advanced demand forecasting with ML
- Supplier portal for direct PO communication
- Customer-facing mobile app for loyalty tracking
- Integration with accounting software (Tally, Zoho)
- Multi-currency and multi-language support

Slide 31 — Thank You / Q&A

- **Title:** Thank You!
- **Contact / GitHub / Demo link**
- **Q&A session**